VIT

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	

Continuous Assessment Test (CAT) – II – September 2026

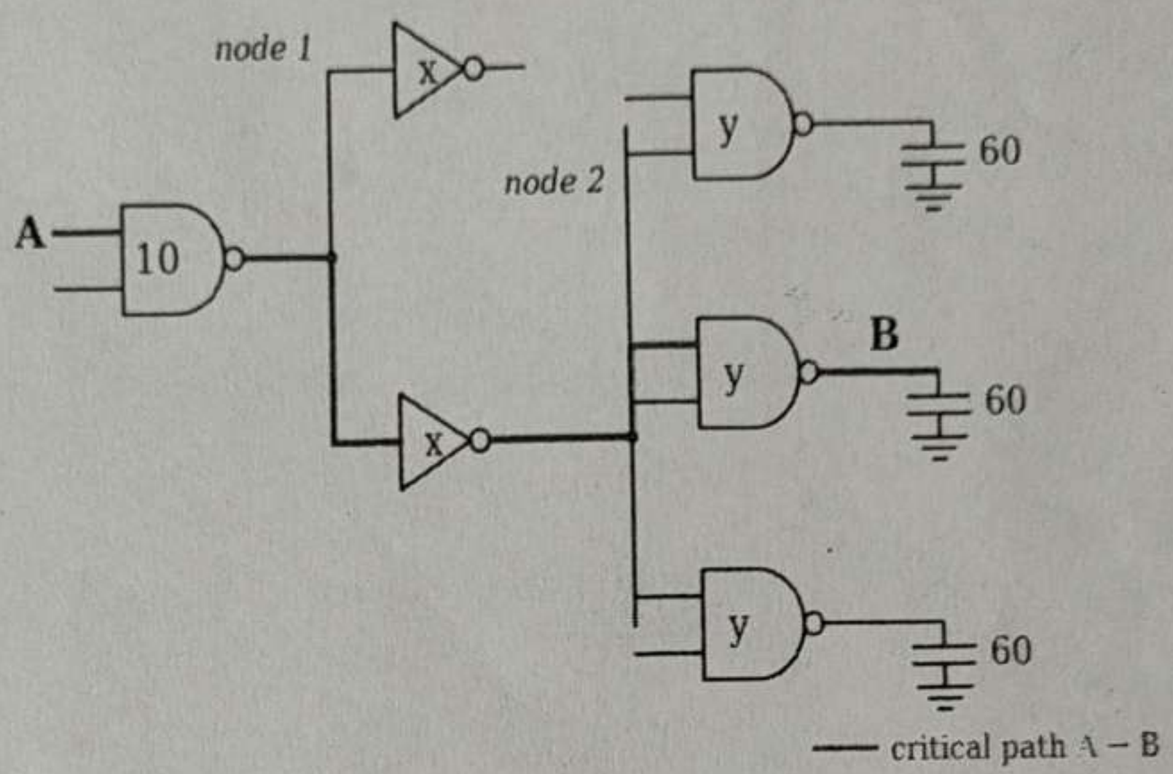
Programme	:	B.Tech – ECE,ECM,BVD	Semester	:	Fall Semester 2026-27
Course Code & Course Title	:	BECE303L & VLSI System Design	Class Number	:	CH2026270102744
Faculty	:	Dr. G. VELMATHI	Slot	:	B1+TB1
Duration	:	90 minutes	Max. Mark		50

Write only your registration number on the question paper in the box provided and do not write other information

Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks
1		A CMOS fabrication facility reports that fabricated chips pass functional testing initially, but some devices exhibit excessive substrate current and eventually undergo destructive latch-up when operated at higher supply voltages. As a process/design engineer, analyze this problem by: a) Comparing the N-well and P-well CMOS fabrication processes, with suitable cross-sectional diagrams and key fabrication steps. (5 Marks) b) Explaining how the parasitic PNPN (SCR) structure responsible for latch-up is inherently formed in CMOS technology. (3 Marks) c) Proposing three process/layout-level techniques to suppress latch-up and explaining why each technique is effective. (2 Marks)	10
2		For the complex gate $Y = [\,(A + B + C) \cdot D\,]'$: (i) Draw the transistor-level CMOS schematic, showing the pull-down and pull-up networks and verifying that they are duals of each other. (2 marks) (ii) using the Euler path theorem, obtain an Euler path for the pull-up graph and for the pull-down graph that yields a COMMON gate ordering, clearly listing the node sequence traversed in each network (3 marks) (iii) Using that ordering, draw the stick diagram so that both the n-diffusion and the p-diffusion strips are single and uninterrupted. Mark the VDD and GND rails, the output node and all contacts. (5 marks)	10

3		Select the gate sizes x and y that give the least delay from A to B in the circuit shown in fig.1  **Fig.1** **(a)** Determine the path logical effort G, the branching effort B, the path electrical effort H and the path effort F. Hence find the optimum stage effort and the minimum delay from A to B. (5 marks) **(b)** Determine the gate sizes x and y, working backwards from the load. Verify your result by checking that the input capacitance of the first gate comes out as 10 units. (5 marks) **(c)** A designer, instead of using your answer, sets x = y = 30 on the argument that "bigger gates are always faster". Compute the resulting A-to-B delay and comment on the outcome. (5 marks)	15
4		Increasing transistor widths in a CMOS block reduces its propagation delay from 120ps to 90 ps but increases its switched capacitance from 30 fF to 45 fF. Supply voltage, switching activity, and operating frequency remain unchanged. Neglect leakage and short-circuit power. a. Calculate the percentage change in delay and switching power. b. Compare the power–delay product before and after sizing.	7
5		Design a CMOS implementation of **[A(B+C)]'** and determine suitable transistor widths so that the worst-case pull-up and pull-down resistances approximately match those of a reference inverter with Wn=W and Wp=2W.	8

**********All the best **********